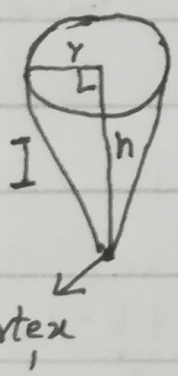
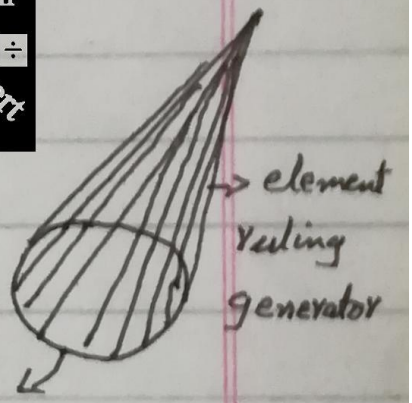
lecture 36

Cone

A surface generated by a straight line passing Through a fixed point and intersecting a fixed plane curve not containing the fixed point is called cone. The given curve is called the directrix and the fixed point is called the vertex of the cone. The straight line is called the generator and any particular position of the generator is called an element or ruling of the cone.



r - radius
h - height
I - Slant height



Right Cone

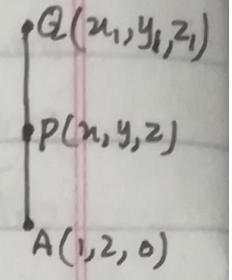
A right cone is a cone with its vertex above the centre of its base.

Right circular cone

A circular con is right circular cone If the centres of its section from a line perpendicular to its base.

Find the equation of cone whose directrix is $x^2 = z$, $y = 3$ and vertex is $A(1,2,0)$

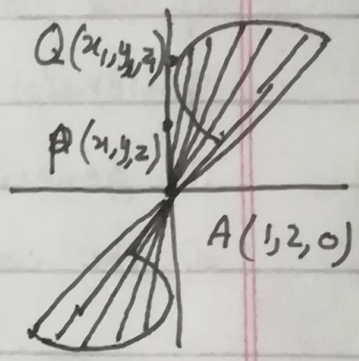
Solution



$x_1^2 = z_1$, $y_1 = 3$

$$\frac{x-1}{x_1-1} = \frac{y-2}{y_1-2} = \frac{z-0}{z_1-0}$$

$$\frac{x-1}{x_1-1} = \frac{y-2}{3-2} = \frac{y-2}{1} = y-2 \qquad x^2 = z,\ y = 3$$



$$\frac{x-1}{x_1-1} = y-2$$

$$x-1 = (y-2)(x_1-1)$$

$$(x_1-1)(y-2) = x-1$$

$$x_1 - 1 = \frac{x-1}{y-2}$$

$$x_1 = \frac{x-1}{y-2} + 1$$

$$x_1 = \frac{x-1+y-2}{y-2}$$

$$x_1 = \frac{x+y-3}{y-2} \quad \text{— I}$$

$$\frac{z}{z_1} = \frac{y-2}{y_1-2}$$

$$\frac{z}{z_1} = \frac{y-2}{3-2}$$

$$\frac{z}{z_1} = \frac{y-2}{1}$$

$$\frac{z}{z_1} = y-2$$

$$(y-2)z_1 = z$$

$$z_1 = \frac{z}{y-2} \quad \text{— II}$$

$x_1^2 = z_1$

$\left(\frac{x+y-3}{y-2}\right)^2 = \frac{z}{y-2}$

$\frac{(x+y-3)^2}{(y-2)^2} = \frac{z}{y-2}$

$(x+y-3)^2 = \frac{z}{y-2} \times (y-2)^2$

$(x+y-3)^2 = z(y-2)$

$[x+y+(-3)]^2 = yz-2z$

formula $(a+b+c)^2 =$

$a^2+b^2+c^2+2ab+2bc+2ca$

$x^2+y^2+(-3)^2+2(x)(y)+2(y)(-3)+2(-3)(x) = yz-2z$

$x^2+y^2+9+2xy-6y-6x-yz+2z=0$ is equation of cone. $x^2+y^2+2xy-yz-6x-6y+2z+9=0$

x - Intercept $y=0$, $z=0$

$x^2+(0)^2+9+2(x)(0)-6(0)-6x-(0)(0)+2(0)=0$

$x^2+0+9+0-0-6x-0+0=0$

$x^2-6x+9=0$, $x^2-3x-3x+9=0$

$x(x-3)-3(x-3)=0$, $(x-3)(x-3)=0$

$x-3=0$, $x-3=0$, $x=3$, $x=3$

y - Intercept $x=0$, $z=0$

$(0)^2+y^2+9+2(0)y-6y-6(0)-y(0)+2(0)=0$

$0+y^2+9+0-6y-0-0+0=0$

$y^2-6y+9=0$

$y^2-3y-3y+9=0$, $y^2-3y-3y+9=0$

$y(y-3)-3(y-3)=0$, $(y-3)(y-3)=0$

$y-3=0$, $y-3=0$

$y=3$ $y=3$

z-Intercept $x=0$, $y=0$

$(0)^2+(0)^2+9+2(0)(0)-6(0)-6(0)-(0)2+2z=0$

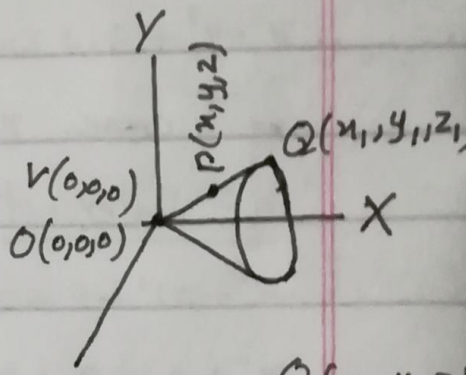
$0+0+9+0-0-0-0+2z=0$

$9+2z=0$, $2z=-9$, $z=-9/2$

Example $z=-9/2$

Find equation of the cone whose directrix is $y^2+z^2=1$, $x=2$ and vertexis $v(0,0,0)$

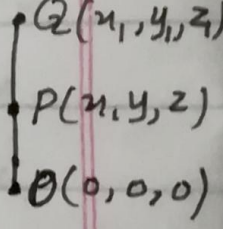
Solution



$y_1^2+z_1^2=1$, $x_1=2$

$\frac{x-0}{x_1-0}=\frac{y-0}{y_1-0}=\frac{z-0}{z_1-0}$

$\frac{x}{x_1}=\frac{y}{y_1}$, $\frac{x}{x_1}=\frac{z}{z_1}$



$\frac{x}{2}=\frac{y}{y_1}$, $\frac{x}{2}=\frac{z}{z_1}$

$xy_1=2y$, $xz_1=2z$

$y_1=\frac{2y}{x}$ — I , $z_1=\frac{2z}{x}$ — II

$\left(\frac{2y}{x}\right)^2+\left(\frac{2z}{x}\right)^2=1$

$\frac{4y^2}{x^2}+\frac{4z^2}{x^2}=1$

$\frac{4y^2+4z^2}{x^2}=1$

$4y^2+4z^2=x^2$ is equation of cone

Sir Naimat ullah Maths Expert

Elliptic Cone

A surface defined by an equation of

The form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = cz^2 \quad (c>0)$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = z^2 \quad (c=1)$ along z-axis

$\frac{y^2}{a^2} + \frac{z^2}{b^2} = x^2$ along x-axis

$\frac{x^2}{a^2} + \frac{z^2}{b^2} = y^2$ along y-axis

Properties

(1) $\frac{x^2}{a^2} + \frac{y^2}{b^2} - z^2 = 0 \quad \because f(x,y,z) = 0$

It is symmetric about xy-plane, yz-plane zx-plane, x-axis, y-axis, z-axis and origin because the power of x, y and z is even

Intercept

x-Intercept $y=0, z=0$

$\frac{x^2}{a^2} + 0 - 0 = 0, \quad x = 0, (0,0,0)$

y-Intercept $x=0, z=0$ Then $y=0, (0,0,0)$

z-Intercept $x=0, y=0$ Then $z=0, (0,0,0)$

Traces

xy-plane $, z=0$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} - z^2 = 0, \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 0$

yz-plane $x=0$

$\frac{y^2}{b^2} - z^2 = 0, \quad \frac{y^2}{b^2} = z^2, \quad y^2 = b^2z^2$

$y = \pm bz$

zx-plane $y=0, \quad \frac{x^2}{a^2} = z^2, \quad x^2 = a^2z^2, \quad x = \pm az$